GOPINATH D

Embedded Systems / Firmware Engineer

gopinathdhanabal596@gmail.com |+919626569152 | linkedin.com/in/gopinath-d

PROFESSIONAL SUMMARY

Electronics and Communication professional with knowledge of embedded systems, C programming, microcontrollers, firmware development, and hardware interfacing. Skilled in debugging, sensor integration, and communication protocols, with a strong interest in developing reliable embedded solutions.

TECHNICAL SKILLS

Programming: Embedded C, C++, Python (basics)

Microcontrollers: ESP32, Arduino (AVR), 8051, microcontroller architecture and interfacing, FreeRTOS/RTOS concepts, Embedded Linux (basics)

Peripherals & Protocols: UART, SPI, I2C, GPIO, ADC, PWM, timers, interrupts, Wi-Fi, Bluetooth, MQTT

Firmware & Tools: Arduino IDE, ESP-IDF (basics), VS Code, Tinker CAD, Git/GitHub, datasheet and schematic analysis

PCB Design: KiCad, Proteus, Cadence (basics), schematic capture, component placement, routing, design rule checks

Hardware & Test: PCB soldering (through-hole & SMD), circuit assembly, oscilloscope, digital multimeter, function generator, component identification, ESD-safe handling

PROFESSIONAL EXPERIENCE

PCB Design Intern Coimbatore

Rehabionics Health Care Pvt. Ltd. [Sep-2025]

- Designed and reviewed PCB layouts from circuit schematics using industry-standard EDA tools for medical electronics.
- Performed component placement, track routing, and design optimization to support signal integrity and manufacturability.
- Verified PCB layouts against schematic netlists and connectivity requirements to identify design errors before fabrication.

Embedded system intern Coimbatore

Emig Litz technologies [July-2025 Sep-2025]

- Gained hands-on experience in embedded systems development, electronic hardware interfacing, and microcontroller-based applications.
- Assisted in hardware integration, circuit testing, troubleshooting, and functional verification of electronic systems.
- Applied embedded systems concepts and automation techniques to develop practical electronics solutions.

PROJECTS

Smart Agriculture Monitoring System (IoT) [2024]

ESP32 | Embedded C | soil moisture and environmental sensors | Wi-Fi | IoT cloud platform | mobile app

- Developed ESP32 firmware for soil moisture, temperature, and humidity monitoring using ADC and I2C sensors.
- Implemented IoT-based automated irrigation using sensor thresholds and Wi-Fi connectivity.
- Integrated a mobile application for real-time monitoring, data visualization, and automated alerts.

Hybrid Transformer-SNN for Rail Squat Detection [2026]

Arduino | vibration sensors | continuous wavelet transforms (CWT) | Spiking Neural Network | Transformer

- Developed a hybrid Transformer-SNN model for railway squat defect detection using vibration data.
- Applied Continuous Wavelet Transform (CWT) for signal processing and time-frequency feature extraction.
- Implemented microcontroller-based data acquisition and evaluated model performance for defect classification.

EDUCATION

Bachelor of Engineering, Electronics and Communication Engineering 2022 - 2026

Info Institute of Engineering, Coimbatore | CGPA: 8.32/ 10

Higher Secondary Certificate (HSC) 2021 - 2022

Zamindar's Higher Secondary School | 71.6%

CERTIFICATIONS & ACHIEVEMENTS

Embedded System Design, Emi glitz Technologies | PCB Designer, Rehabionics Health Care Pvt
Laser Source Repair Trainee, Laser Experts India LLP | Participant, Smart India Hackathon. Ltd.